

OSLAB PRACTICAL-6 PROGRAMS & OUTPUT

Date: 09-09-26

Program-1 & OUTPUT:

```
sainath@Sainath:~/OSLAB/practical_6$ cat program1.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/stat.h>
#include <string.h>

#define FIFO1 "client_to_server"
#define FIFO2 "server_to_client"

int main() {
    char buffer[100];
    char response[150];

    // Create named pipes
    mkfifo(FIFO1, 0666);
    mkfifo(FIFO2, 0666);

    printf("Server waiting for client...\n");

    // Open FIFOs
    int read_fd = open(FIFO1, O_RDONLY);
    int write_fd = open(FIFO2, O_WRONLY);

    // Read message from client
    read(read_fd, buffer, sizeof(buffer));

    printf("Server received: %s\n", buffer);

    // Process and prepare response
    sprintf(response, "Server processed: %s", buffer);

    // Send response to client
    write(write_fd, response, strlen(response) + 1);

    close(read_fd);
    close(write_fd);

    // Remove FIFOs
    unlink(FIFO1);
    unlink(FIFO2);

    return 0;
}
```

```
sainath@Sainath:~/OSLAB/practical_6$ gedit program1.c
sainath@Sainath:~/OSLAB/practical_6$ gcc program1.c -o program1
sainath@Sainath:~/OSLAB/practical_6$ ./program1
Server waiting for client...
```

Program-2 & OUTPUT:

```
sainath@Sainath:~/OSLAB/practical_6$ cat program2.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#include <string.h>

#define FIFO1 "client_to_server"
#define FIFO2 "server_to_client"

int main() {
    char message[100];
    char response[100];

    printf("Enter message: ");
    fgets(message, sizeof(message), stdin);

    // Open FIFOs
    int write_fd = open(FIFO1, O_WRONLY);
    int read_fd = open(FIFO2, O_RDONLY);

    // Send message to server
    write(write_fd, message, strlen(message) + 1);

    // Receive response
    read(read_fd, response, sizeof(response));

    printf("Server response: %s\n", response);

    close(write_fd);
    close(read_fd);

    return 0;
}
```

```
sainath@Sainath:~/OSLAB/practical_6$ gedit program2.c
sainath@Sainath:~/OSLAB/practical_6$ gcc program2.c -o program2
sainath@Sainath:~/OSLAB/practical_6$ ./program2
Enter message: Hello Server
```

History:

```
367 cd OSLAB
368 ls
369 mkdir practical_6
370 cd practical_6
371 gedit program1.c
372 gcc program1.c -o program1
373 gedit program1.c
374 gcc program1.c -o program1
375 ./program1
376 gedit program2.c
377 gcc program2.c -o program2
378 ./program2
379 cat program1.c
380 cat program2.c
381 history
```